

RECOVERING THE INITIAL VALUES FROM THE SOLUTIONS OF THE FRACTIONAL WAVE EQUATION ON THE LIGHT CONE TRACE

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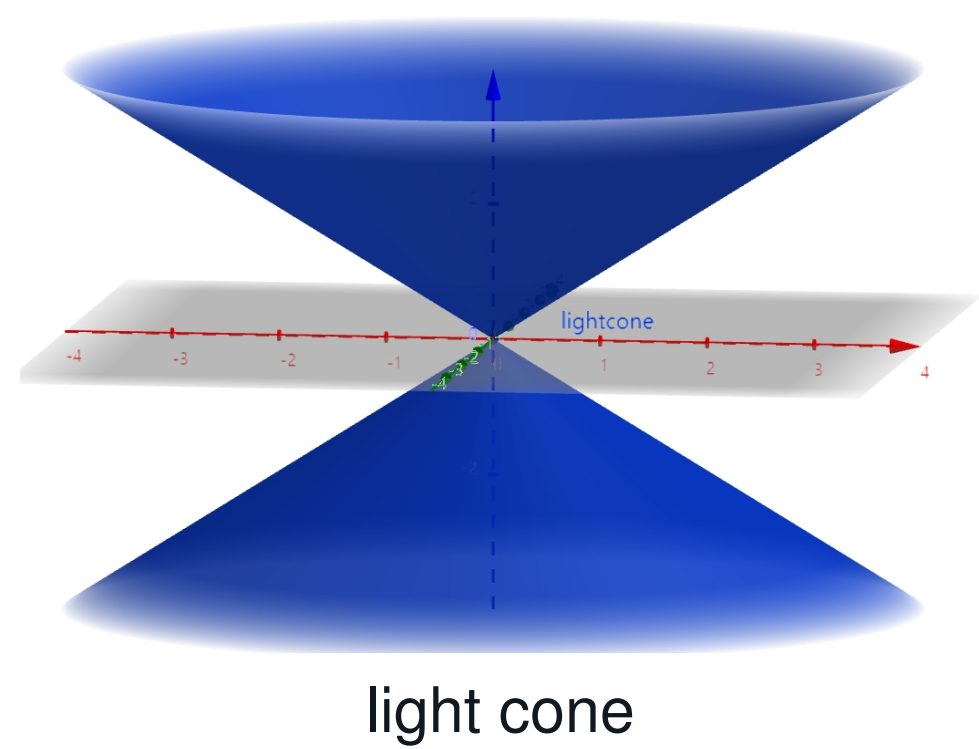
Abstract

We reconstruct the initial functions from the trace of solution of an initial value problem for the wave equation on the light cone. When the spatial dimension is odd, Rakesh and Yuan have already provide the inverting the operator trace the solution on the light cone from the initial function [4]. We generalize their work to the fractional wave equation and all dimensions. That is, we present how to reconstruct the initial functions from the solution of the fractional wave equation on the light cone.

Introduction

light cone

- (a) The path in all directions through which the light can travel along space-time.
- (b) Satisfies the wave equation because light is a wave.
- (c) Can express light cone as $t = \pm|\mathbf{x}|$ and here we deal with $t = |\mathbf{x}|$ case.



light cone

$\left\{ \begin{array}{l} \text{time axis : one second} \\ \text{spatial axes : 300,000 km} \end{array} \right.$

The fractional wave equation

$$\begin{aligned} D_t^\alpha p(\mathbf{x}, t; \alpha) &= -(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} p(\mathbf{x}, t; \alpha), \\ (\mathbf{x}, t) &= (x_1, x_2, t) \in \mathbb{R}^2 \times (0, \infty), \quad 1 < \alpha \leq 2 \end{aligned} \quad (1)$$

- The Riesz space-fractional derivative of order α

$$\mathcal{F}(-(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} f)(\xi) := -|\xi|^\alpha (\mathcal{F}f)(\xi)$$

where $\mathcal{F}$: the Fourier transform

- The Caputo time-fractional derivative of order α

$$(D^\alpha h)(t) := (I^{m-\alpha} h^{(m)})(t), \quad m-1 < \alpha \leq m, \quad m \in \mathbb{N},$$

where I^α : the Riemann-Liouville fractional integral

$$(I^\alpha h)(t) := \begin{cases} \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} h(\tau) d\tau, & \text{if } \alpha > 0, \\ h(t), & \text{if } \alpha = 0 \end{cases}$$

- The Mittag-Leffler function

$$E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(1+\alpha k)}, \quad \operatorname{Re}(\alpha) > 0, \quad z \in \mathbb{C}. \quad (2)$$

Preliminary

- The Fourier series

$$f(r\boldsymbol{\theta}) = \sum_{m=-\infty}^{\infty} f_m(r) Y_m(\boldsymbol{\theta}), \quad (3)$$

where $Y_m(\boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi}} e^{im\theta}$, $(r, \boldsymbol{\theta}) \in [0, \infty) \times \mathbb{S}^1$ with $\boldsymbol{\theta} = (\cos \theta, \sin \theta)$.

- The Mellin transform

$$\mathcal{M}f(s) = \int_0^{\infty} x^{s-1} f(x) dx, \quad s \in \mathbb{C}. \quad (4)$$

- Property of Mellin transform

$$\mathcal{M}(f \times g) = \mathcal{M}f \cdot \mathcal{M}g,$$

$$\text{where } f \times g(t) = \int_0^{\infty} f(\tau) g\left(\frac{t}{\tau}\right) \frac{d\tau}{\tau}.$$

Suppose that

$$f(x) = \begin{cases} O(x^{-a-\epsilon}), & \text{as } x \rightarrow 0^+, \\ O(x^{-b+\epsilon}), & \text{as } x \rightarrow +\infty, \end{cases} \quad \epsilon > 0, \quad a < b$$

- The inversion formula for the Mellin transform

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} t^{-s} \mathcal{M}f(s) ds, \quad a < c < b. \quad (5)$$

Main results

Suppose $f \in C^\infty(\mathbb{R}^2)$ have compact support.

Then the solution of the fractional wave equation (1) with the initial condition

$$p(\mathbf{x}, 0; \alpha) = f(\mathbf{x}), \quad \text{and} \quad D_t p(\mathbf{x}, t; \alpha)|_{t=0} = 0,$$

is

$$p(\mathbf{x}, t; \alpha) = \frac{1}{(2\pi)^2} \int e^{i\xi \cdot \mathbf{x}} E_\alpha(-|\xi|^\alpha t^\alpha) \mathcal{F}f(\xi) d\xi, \quad (6)$$

- Define the linear operator $\mathcal{W}_\alpha$

$$\mathcal{W}_\alpha f(\mathbf{x}) = p(\mathbf{x}, |\mathbf{x}|; \alpha). \quad (7)$$

- **Goal** : **reconstruction $f(\mathbf{x})$ from $p(\mathbf{x}, |\mathbf{x}|; \alpha)$**

- **Strategy**

$$\mathcal{W}_\alpha f \mapsto (\mathcal{W}_\alpha f)_m \mapsto (\mathcal{F}f)_m \mapsto \mathcal{F}f \mapsto f$$

Theorem 1

For $f \in C^\infty(\mathbb{R}^2)$ have compact support,

$$(\mathcal{W}_\alpha f)_m(r) = \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha r^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi r) d\rho_\xi,$$

where $J_\nu(\cdot)$: Bessel function (of the first kind) of order ν .

Proof.

$$\begin{aligned} \mathcal{W}_\alpha f(\mathbf{x}) &= \frac{1}{(2\pi)^2} \int_{\mathbb{S}^1} \int_0^\infty e^{i\rho_\xi \boldsymbol{\theta}_\xi \cdot \mathbf{x}} E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) \mathcal{F}f(\rho_\xi \boldsymbol{\theta}_\xi) \rho_\xi d\rho_\xi dS(\boldsymbol{\theta}_\xi) \\ &= \frac{1}{(2\pi)^2} \sum_{m=-\infty}^{\infty} \int_0^\infty E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi \left(\int_{\mathbb{S}^1} e^{i\rho_\xi \boldsymbol{\theta}_\xi \cdot \mathbf{x}} Y_m(\boldsymbol{\theta}_\xi) dS(\boldsymbol{\theta}_\xi) \right) d\rho_\xi \\ &= \sum_{m=-\infty}^{\infty} \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi |\mathbf{x}|) d\rho_\xi Y_m\left(\frac{\mathbf{x}}{|\mathbf{x}|}\right), \end{aligned}$$

The Funk-Hecke theorem in [2, p.198] :

$$\int_{\mathbb{S}^1} e^{i\rho_\xi |\mathbf{x}| \boldsymbol{\theta}_\xi \cdot \frac{\mathbf{x}}{|\mathbf{x}|}} Y_m(\boldsymbol{\theta}_\xi) dS(\boldsymbol{\theta}_\xi) = (2\pi) i^m J_m(\rho_\xi |\mathbf{x}|) Y_m\left(\frac{\mathbf{x}}{|\mathbf{x}|}\right).$$

□

- To Use the mellin transform

$$r^{-2} (\mathcal{W}_\alpha f)_m(r^{-1}) = \frac{(-1)^{|m|} i^m}{2\pi} (\mathcal{F}f)_m \times E(r; \alpha, m),$$

where $E(r; \alpha, m) = E_\alpha(-r^{-\alpha}) J_{|m|}(r^{-1}) r^{-2}$, beacuse $J_m = (-1)^{|m|} J_{|m|}$.

Theorem 2

Let $f \in C^\infty(\mathbb{R}^2)$ have compact support.

For $\frac{3}{2} + \epsilon < c < 2 + |m| - \epsilon$ with $\epsilon < \frac{1}{4}$, we have

$$(\mathcal{F}f)_m(r) = i^{m-1} \int_{c-i\infty}^{c+i\infty} r^{-s} \frac{\mathcal{M}[(\cdot)^{-2} (\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

Proof.

$$\begin{aligned} \mathcal{M}[(\cdot)^{-2} (\mathcal{W}_\alpha f)_m(\cdot^{-1})](s) &= (-1)^{|m|} i^m (2\pi)^{-1} \mathcal{M}[(\mathcal{F}f)_m \times E](s; \alpha, m) \\ &= (-1)^{|m|} i^m (2\pi)^{-1} \mathcal{M}((\mathcal{F}f)_m)(s) \cdot \mathcal{M}(E)(s; \alpha, m). \end{aligned}$$

Taking the Mellin inverse formula (5) of

$$\mathcal{M}((\mathcal{F}f)_m)(s) = \frac{i^m (2\pi) \mathcal{M}[(\cdot)^{-2} (\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)},$$

we have

$$(\mathcal{F}f)_m(t) = i^{m-1} \int_{c-i\infty}^{c+i\infty} t^{-s} \frac{\mathcal{M}[(\cdot)^{-2} (\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

□

Conclusion and Summary

Here is our summarization how to recover f from $\mathcal{W}_\alpha f$ as follows:

1. We can find $(\mathcal{W}_\alpha f)_{lk}$ using the spherical harmonics expansions of $\mathcal{W}_\alpha f$.
2. We can find the relation between $(\mathcal{F}f)_m$ and $(\mathcal{W}_\alpha f)_m$ (see Thorem 1)
3. Taking the Mellin inversion formula of $\mathcal{M}[(\cdot)^{-n} (\mathcal{W}_\alpha f)_{lk}(\cdot^{-1})]$, we can find $(\mathcal{F}f)_{lk}$ (see Theorem 2).
4. We recover f from $(\mathcal{F}f)_{lk}$ through $\mathcal{F}f$.

Remark

In three dimensions, f can be reconstructed from $\mathcal{W}_\alpha f$ in the same way as reconstructed in two dimensions.

$$\text{with } Y_{lk}(\theta, \varphi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-|k|)!}{(l+|k|)!}} P_l^{|k|}(\cos \theta) e^{ik\varphi}, \quad k = -l, \dots, l, \quad l = 0, 1, 2, \dots$$

In all dimensions, we can reconstruct f from $\mathcal{W}_\alpha f$.

References

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